Carlos Diogo

Software Engineer

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Education

Faculty of Engineering, University of Porto (FEUP) Sep 2024 – Jun 2027
B.S. in Informatics and Computing Engineering *Porto, Portugal*

Colégio de Gaia Sep 2021 – Jun 2024
High School Diploma | Informática e Tecnologias Multimédia *Vila Nova de Gaia*

- GPA: 19.7/20
- Self-taught programming during high school

Experience

Academic Tutor Oct 2025 – Jul 2026
Faculty of Engineering, University of Porto *Porto, Portugal*

- Created clear learning materials and support resources for Calculus I and II

Power Platform Software Developer Intern Dec 2023 – May 2024
APR - Technology Solutions *Ermesinde, Porto, Portugal (Remote)*

- Built a Microsoft Power Apps solution that simplified how users record, organise and monitor expenses
- Improved the user experience and streamlined financial workflows through practical low-code development

Power Platform Software Developer Intern Jun 2023 – Nov 2023
bwd Digital Transformation *Porto, Portugal (Hybrid)*

- Helped develop a Microsoft Power Apps solution for an international company, improving business workflows and operational efficiency
- Designed automated processes with Microsoft Power Automate to reduce manual work and improve process consistency

Projects

Ark (github.com/rodrgds/ark · CodeSpring Hackathon, ISEP · 3rd place) May 2026

- Built an offline-first survival companion for iOS and Android combining downloadable offline maps, a searchable knowledge library, protected notes and documents with encrypted backups, and an on-device AI assistant that searches the user's own sources
- Added field tools (compass, barometer, level, weather cache, GPX recording) using Expo and React Native, SQLite full-text search, and native modules for routing, OCR and ZIM reading

Compiler Register Allocation (GitHub · 3-person team · Algorithm Design) May 2026 – May 2026

- Developed a global register allocator that merges live ranges into webs, builds an interference graph and assigns registers under a fixed limit
- Implemented baseline colouring, controlled spilling, live-range splitting, DSatur ordering, graph-class shortcuts and guarded branch-and-bound refinement
- Added free-split recovery, marker-preserving output, DOT graph exports, deterministic integration tests and clear visualisations

ARC Gym (3-person team · Web Languages and Technologies) Mar 2026 – Jun 2026

- Co-developed a gym management platform with strict PHP, SQLite, semantic HTML, layered CSS and vanilla JavaScript/AJAX, without application or CSS frameworks
- Implemented class registration and waitlists, trainer, member and administrator workflows, personal training, equipment reservations, memberships, notifications and analytics
- Added QR check-in, physical membership cards, a Bearer-authenticated JSON/XML REST API, migrations, audit logs, backups, session controls and security hardening

Scientific Review Assignment (GitHub · 3-person team · Algorithm Design) Mar 2026 – May 2026

- Modelled reviewer-to-paper assignment as a capacitated bipartite flow network with expertise, workload and reviewer-capacity constraints
- Implemented Ford-Fulkerson and Edmonds-Karp in C++, with batch and interactive workflows, CSV parsing, assignment exports and reviewer-absence analysis
- Documented graph construction, complexity and algorithm trade-offs with Doxygen and Graphviz visualisations

INFOCUIDA (Software Engineering course, FEUP) Feb 2026 – Jun 2026

- Built a mobile platform that gives informal caregivers learning resources (tutorials, videos, legal guidance, FAQs, PDF exports), moderated peer-support forums, and professional profiles with job listings and applications
- Supported accessibility with guided onboarding, simplified navigation and customisable settings; built with Flutter, Dart and Firebase (Auth, Firestore, Storage, Cloud Messaging)

Ninjix (4-person team · LCOM) Feb 2026 – Jun 2026

- Co-developed a pixel-art tower-defence game and lightweight C game engine for MINIX, driven by timer, keyboard, mouse, graphics and serial-port interrupts
- Implemented a 60 Hz simulation and rendering loop with levels, waves, tower upgrades, enemy abilities, projectile pools, menus, profiling and PNG-to-XPM tooling
- Added two-player UART multiplayer with attacker and defender roles, binary synchronisation, handshakes, retransmission and shared game-state events

TankTussle (GitHub · 3-person team · Software Design and Testing) Sep 2025 – Jan 2026

- Co-developed a TankTrouble-inspired 2D tank combat game in Java with Lanterna, supporting local 1v1 multiplayer and single-player AI
- Implemented bouncing projectile physics, ricochets, destructible and indestructible walls, multiple maps, a map editor, menus, settings, sound, music and particles
- Structured the game around MVC, state and event-driven architecture, design patterns, Gradle automation and a comprehensive testing toolchain

NatuReal (GitHub · CodeSpring Hackathon, ISEP · 1st place) Apr 2024

- Built a bird-watching web application with daily challenges, friends, rankings and a photo publication feed during a three-day hackathon
- Developed the backend in PHP and MySQL/MariaDB with JWT authentication and PHPMailer, and the frontend with Tailwind CSS, DaisyUI and jQuery

FisioFinder (Professional Aptitude Project, Colégio de Gaia) Nov 2023 – Jun 2024

- Created a healthcare platform for physical medicine and rehabilitation with a mobile app, website and administration portal: appointment and treatment scheduling, medication reminders, a caregiver view, and an interactive clinic map
- Added accessibility features such as voice-based forms and text-to-speech; built with Flutter, Dart, Supabase, PostgreSQL, PHP and the Google Maps API

Honors & Awards

- **1st place**, CodeSpring Hackathon 2024 (ISEP) — NatuReal, bird-watching web application
- **3rd place**, CodeSpring Hackathon 2026 (ISEP) — Ark, offline-first survival companion
- High School Diploma, Informática e Tecnologias Multimédia, Colégio de Gaia — GPA: 19.7/20

Skills

- **Languages:** Dart, TypeScript, JavaScript, C/C++, Java, PHP, Python, SQL
- **Frameworks & Tools:** Flutter, React Native, Expo, Tailwind CSS, DaisyUI, jQuery, Microsoft Power Apps, Power Automate, Gradle, JUnit
- **Infrastructure:** Firebase, Supabase, PostgreSQL, MySQL/MariaDB, SQLite